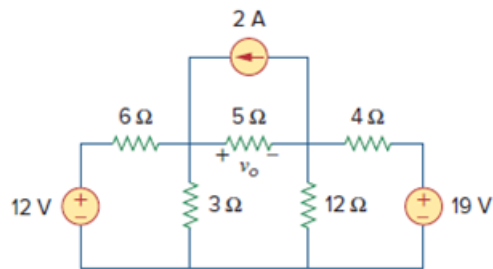


Problem

Determine v_o in the circuit of Fig. 4.80 using the superposition principle.

Figure 4.80:



Step-by-step solution

Step 1 of 12

Let $V_o = V_{o1} + V_{o2} + V_{o3}$

Where V_{o1} , V_{o2} & V_{o3} are due to, 2A, 12V & 19V sources respectively.

Step 2 of 12

For V_{o1} , consider the circuit below;

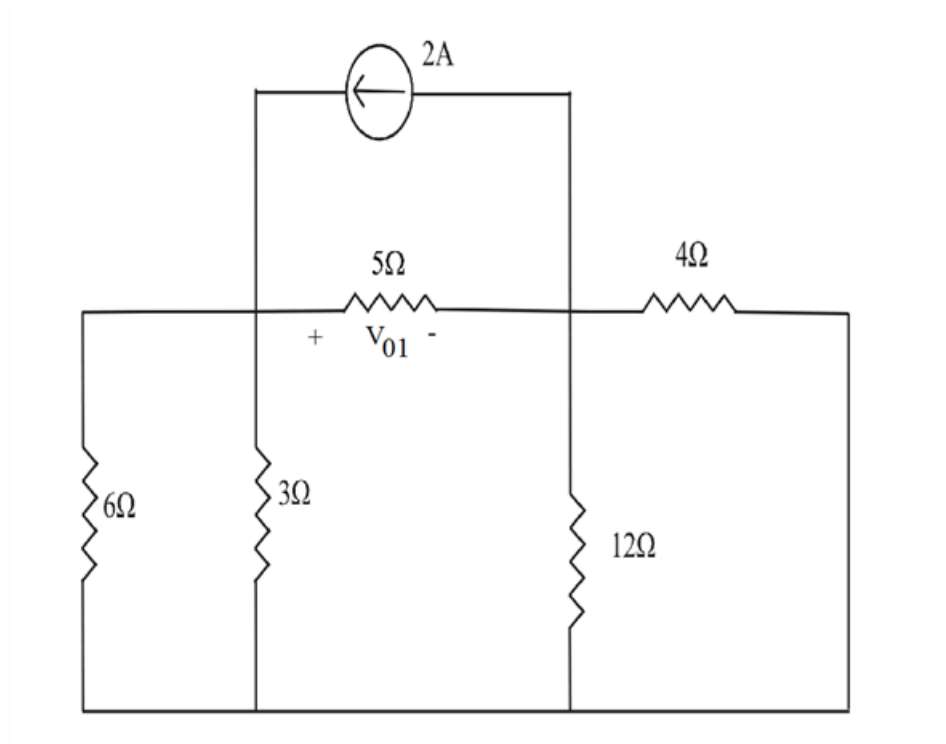


Fig – 1(a)

Step 3 of 12

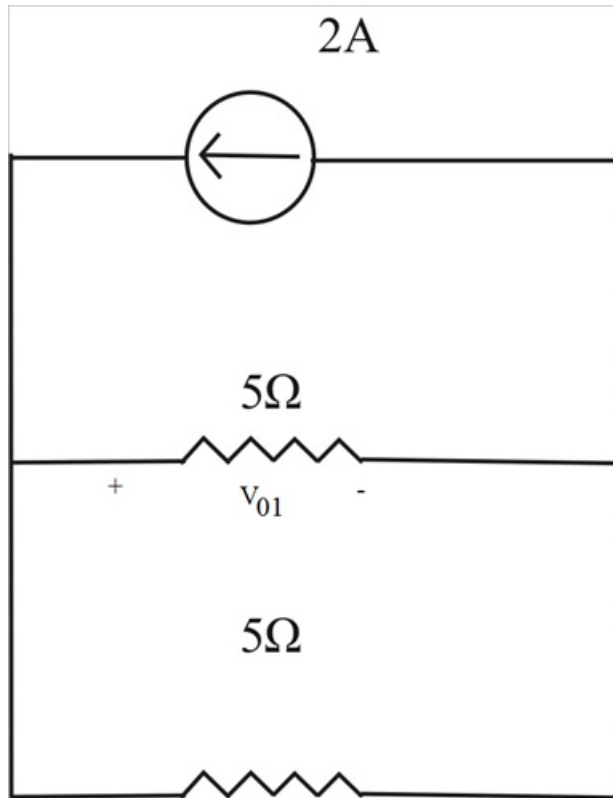


Fig – 1(b)

From Fig-1(a)

$$6\Omega \parallel 3\Omega = \frac{6 \times 3}{6 + 3}$$

$$6\Omega \parallel 3\Omega = 2\Omega$$

$$4\Omega \parallel 12\Omega = \frac{4 \times 12}{4 + 12}$$

$$4\Omega \parallel 12\Omega = 3\Omega$$

We know that $V = IR$ (1)

From Fig -1(b)

$$I = 2A \text{ and}$$

$$R = \frac{5 \times 5}{5 + 5}$$

$$R = 2.5\Omega$$

Therefore,

$$V_{01} = 2 \times 2.5 \text{ (From equation-(1))}$$

$$V_{01} = 5V$$

For V_{02} , consider the circuit below;

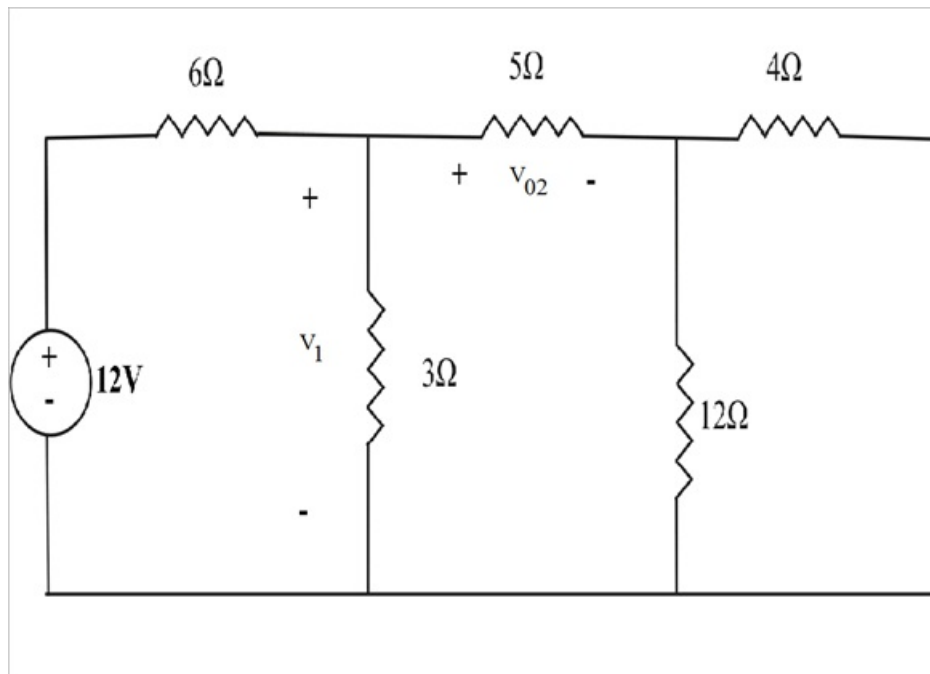


Fig – 1(c)

Step 5 of 12

Fig – 1(c)

$$4\Omega \parallel 12\Omega = \frac{4 \times 12}{4 + 12}$$

$$4\Omega \parallel 12\Omega = 3\Omega$$

$$3\Omega \parallel 8\Omega = \frac{24}{11}$$

Step 6 of 12

By using voltage division rule we get,

$$V_1 = 12 \times \left[\frac{\left(\frac{24}{11} \right)}{6 + \frac{24}{11}} \right]$$

$$V_1 = \frac{16}{5}$$

$$V_{02} = \left(\frac{5}{5+3} \right) V_1 \quad (\text{By using voltage division rule})$$

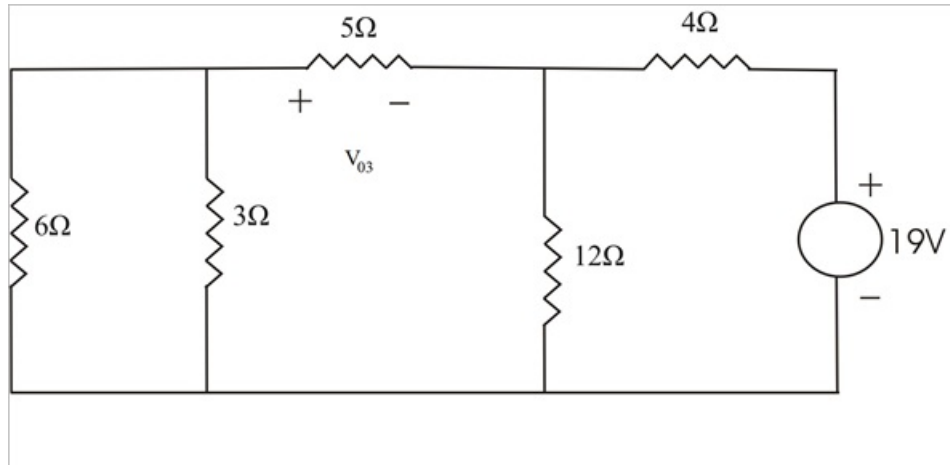
$$V_{02} = \left(\frac{5}{8} \right) V_1$$

$$V_{02} = \frac{5}{8} \times \frac{16}{5}$$

$$V_{02} = 2 \text{ V}$$

Step 7 of 12

For V_{03} , consider the circuit below;



Step 8 of 12

The resistors 6Ω and 3Ω are in parallel,

$$6\Omega \parallel 3\Omega = \frac{6 \times 3}{6 + 3}$$

$$6\Omega \parallel 3\Omega = 2\Omega$$

Step 9 of 12

Therefore the circuit is transformed as

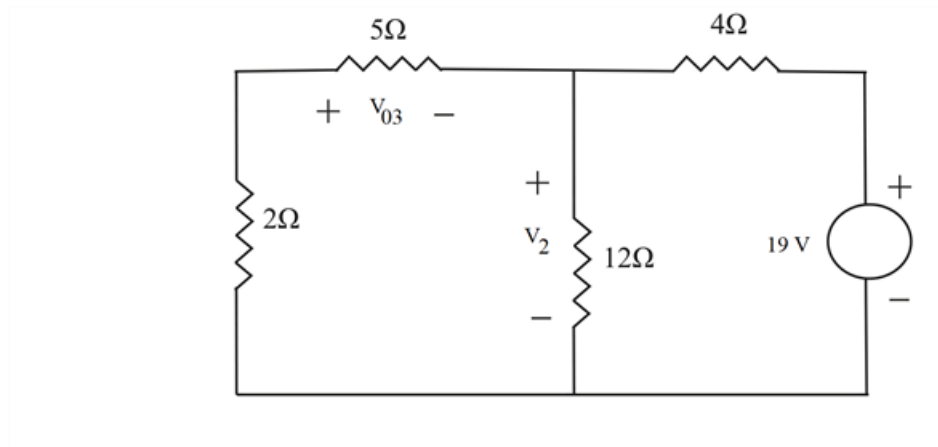


Fig-1(d)

Step 10 of 12

From Fig –1(d)

$$7\Omega \parallel 12\Omega = \left(\frac{84}{19}\right)\Omega$$

By using voltage division rule we get,

$$V_2 = 19 \times \left[\frac{\left(\frac{84}{19}\right)}{4 + \frac{84}{19}} \right]$$

$$V_2 = 9.975$$

Step 11 of 12

$$V_{03} = \left(\frac{5}{5+2}\right)(-V_2) \text{ (By using voltage division rule)}$$

$$V_{03} = \left(\frac{-5}{7}\right)V_2$$

$$V_{03} = \left(\frac{-5}{7}\right)9.975$$

$$V_{03} = -7.125 \text{ V}$$

Step 12 of 12

Therefore Total voltage V_o in the above circuit is;

$$V_o = V_{01} + V_{02} + V_{03}$$

$$V_o = 5 + 2 - 7.125$$

$$V_o = -0.125 \text{ V}$$